Reg. No. | | | | | | | | | | | [14IS43]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE-2016

PROGRAMMING PARADIGMS

Time:3 Hours Maximum Marks:70

Instructions:

1. Instructions: Answer any **FIVE** full questions choosing at least **TWO** questions from each Part.
2. Any missing data may be suitably assumed

PART – A

1.	a.	Explain the reasons for studying programming language concepts.	05
	b.	What is Writability? Describe the important characteristics that influence the writability.	06
	c.	What are programming environments? Explain their role in software development.	03
2.	a.	What is lexical analyser? Explain three methods of building lexical analyser.	05
	b.	Write and explain a state diagrams to recognise names and integers.	03
	c.	What is parsing? Explain two different parsers.	06
3.	a.	What are variables? Explain static attributes of variables.	06
	b.	Which are static languages that support nested subprograms?	04
	c.	Explain different slicing operations in arrays.	04
4.	a.	What are side effects? Suggest any two solutions to the problem operand evaluation and side effects.	05
	b.	Define narrowing and widening conversions with examples.	03
	c.	What are guarded commands? Explain their operation.	06

PART – B

5.	a.	Explain different implementation models of parameter passing.	10
	b.	Define static chain, static depth, nesting depth, chain offset.	04
6.	a.	What is abstraction? Explain it is importance in programming languages.	04
	b.	Explain abstract data types in Ada.	06
	c.	What are parameterized ADT? How are they implemented in C++?	04
7.	a.	Explain the concepts of inheritance and dynamic binding.	06
	b.	What are the uses and problems with multiple inheritances? Explain.	04
	c.	Compare dynamic binding in C++ and java.	04
8.	a.	Explain different categories of concurrency.	04
	b.	What are exceptions? Discuss the different possibilities of exception handling.	06
	c.	Discuss the java features for exception handling.	04